Docker Usage

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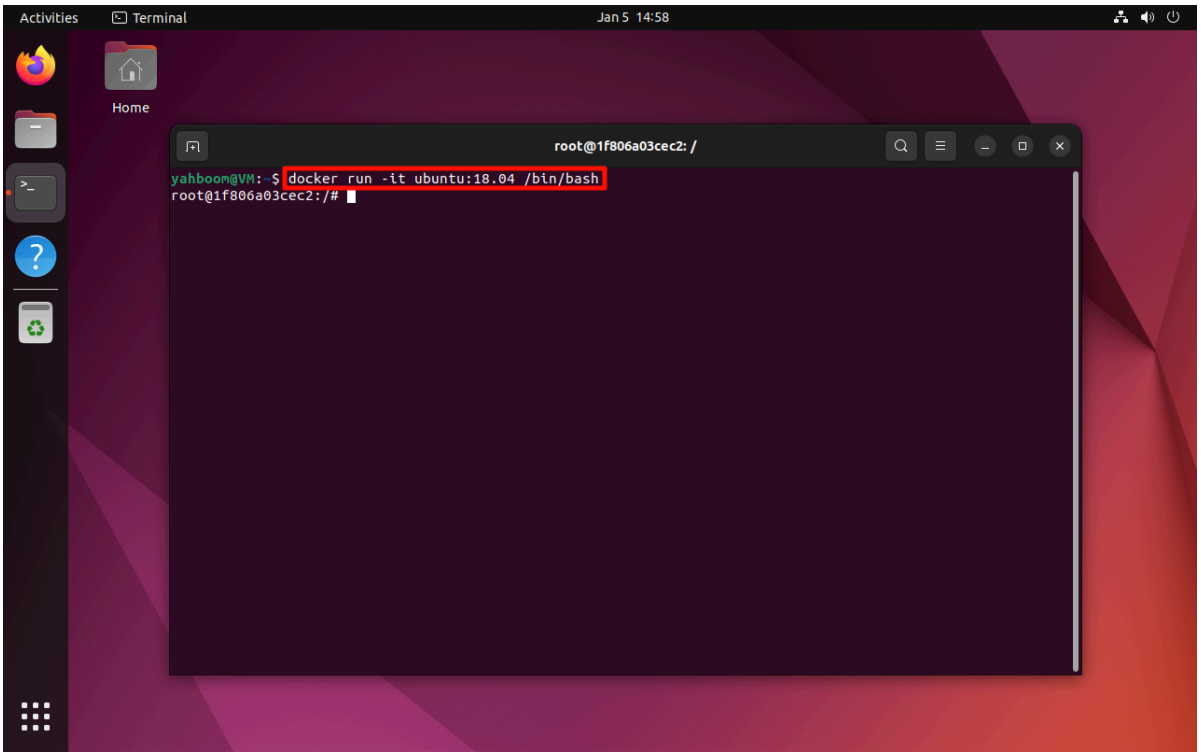
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> **Historical Demonstration:** This section shows building ROS 1 Melodic in an `ubuntu:18.04` container, and is only intended for understanding the process of installing inside a container and committing an image. Both Ubuntu 18.04 and ROS 1 Melodic have reached end of regular support; this process is not an installation guide for the current Ubuntu 24.04 / ROS 2 Jazzy, and the repositories and dependencies may no longer be available.

1. Start the Image

Start the `ubuntu:18.04` image in interactive mode:

```
docker run -it ubuntu:18.04 /bin/bash
```



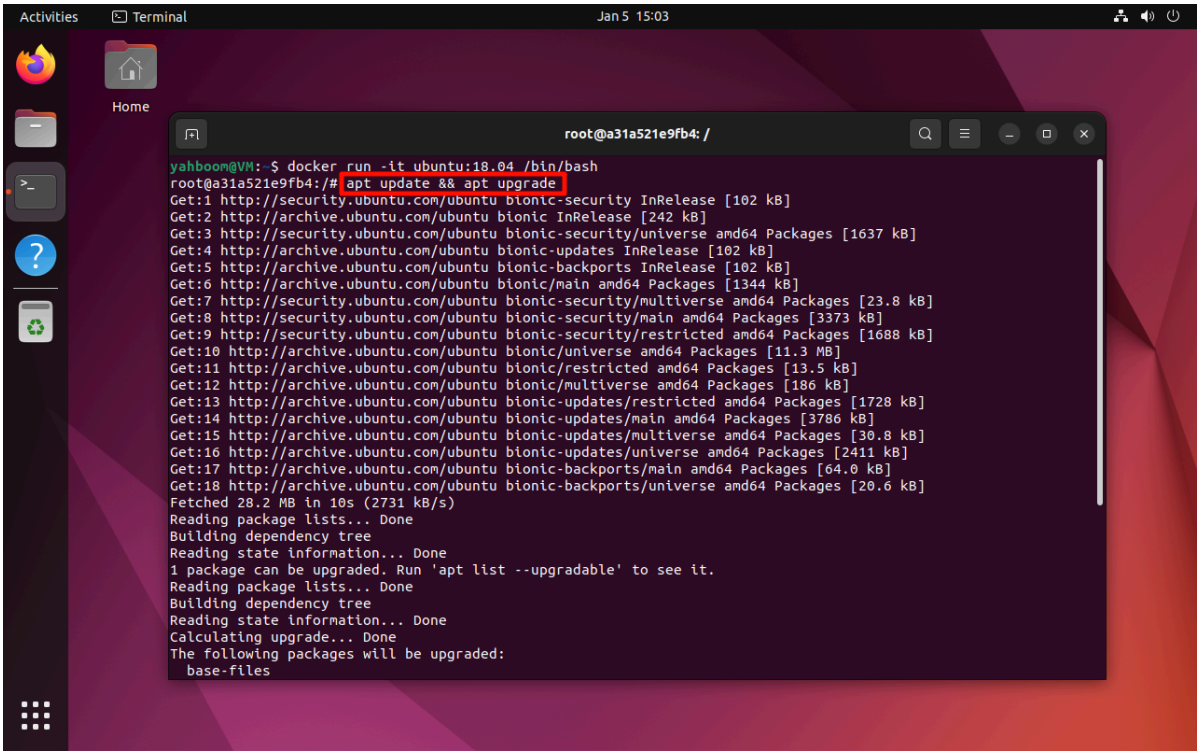
2. ROS Environment Construction

Administrator privileges are not required inside Docker. If the ROS environment is installed on the host, sudo needs to be added to the commands.

2.1. Update System Software

Installing the ROS 1 environment generally requires ensuring that the system software package is up to date:

```
apt update && apt upgrade
```

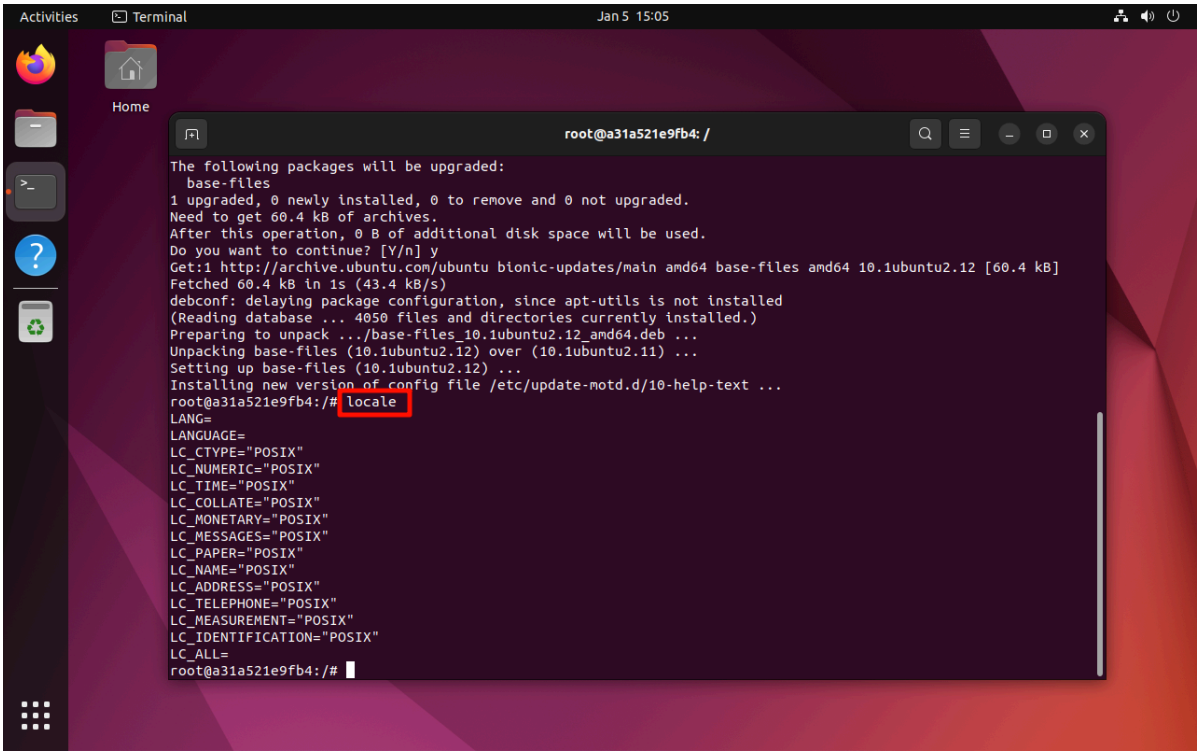


2.2. Determine the Language Environment

Make sure the system language environment supports UTF-8!

2.2.1. Verify the System Environment

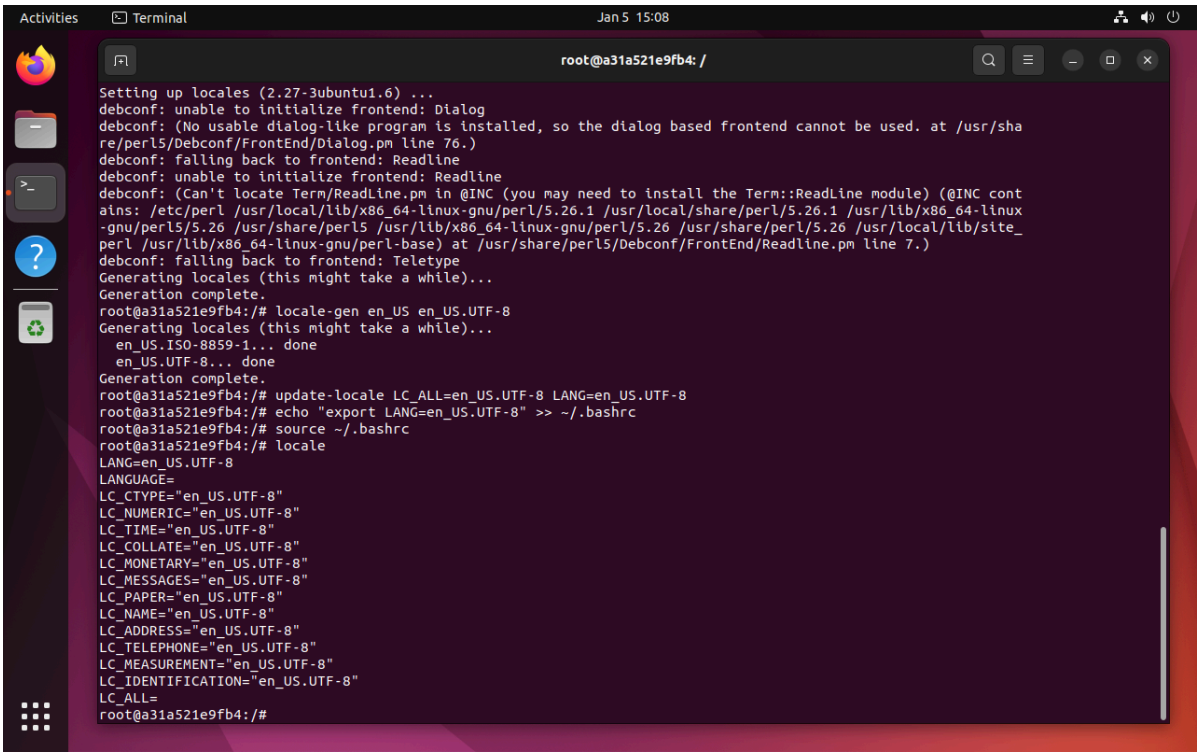
```
locale
```



2.2.2. Set the UTF-8 Environment

If the system language does not support the UTF-8 environment, enter the following commands in the terminal line by line:

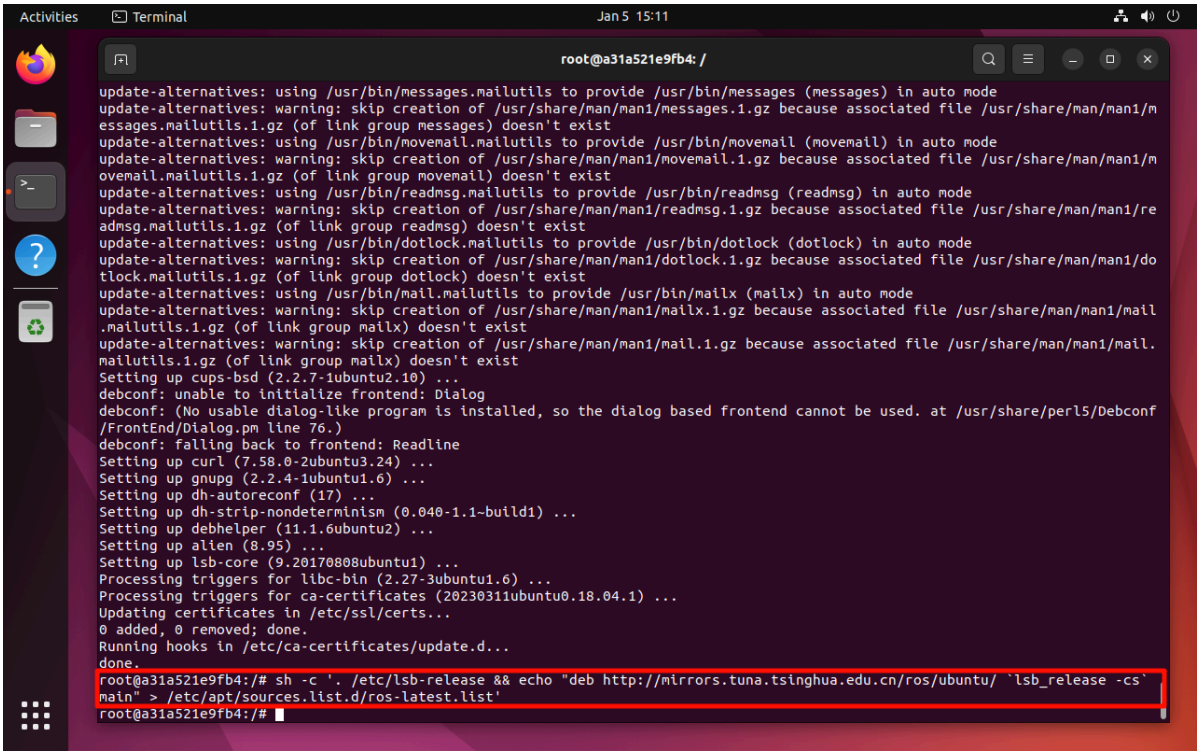
```
apt install locales
locale-gen en_US en_US.UTF-8
update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
echo "export LANG=en_US.UTF-8" >> ~/.bashrc
source ~/.bashrc
```



2.3. Set the Software Source

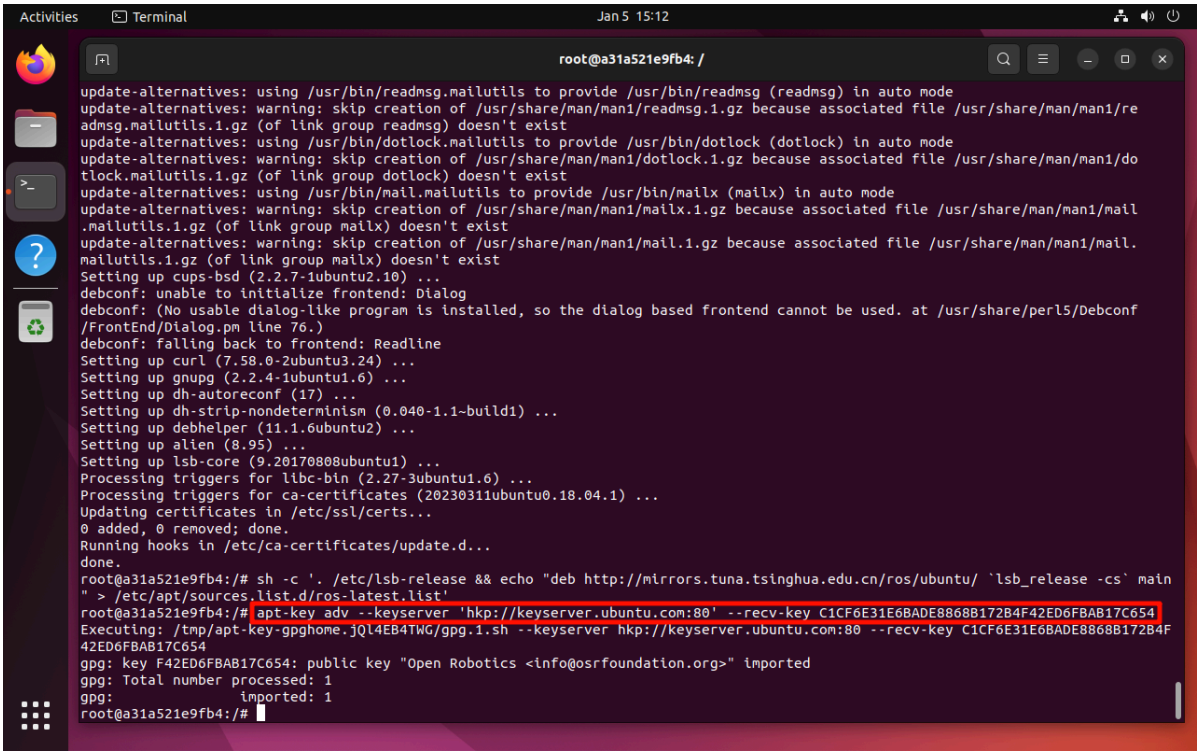
```
apt install lsb-core
```

```
sh -c '. /etc/lsb-release && echo "deb http://mirrors.tuna.tsinghua.edu.cn/ros/ubuntu/ `lsb_release -cs` main" > /etc/apt/sources.list.d/ros-latest.list'
```



2.4. Set the Key

```
apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key
C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
```

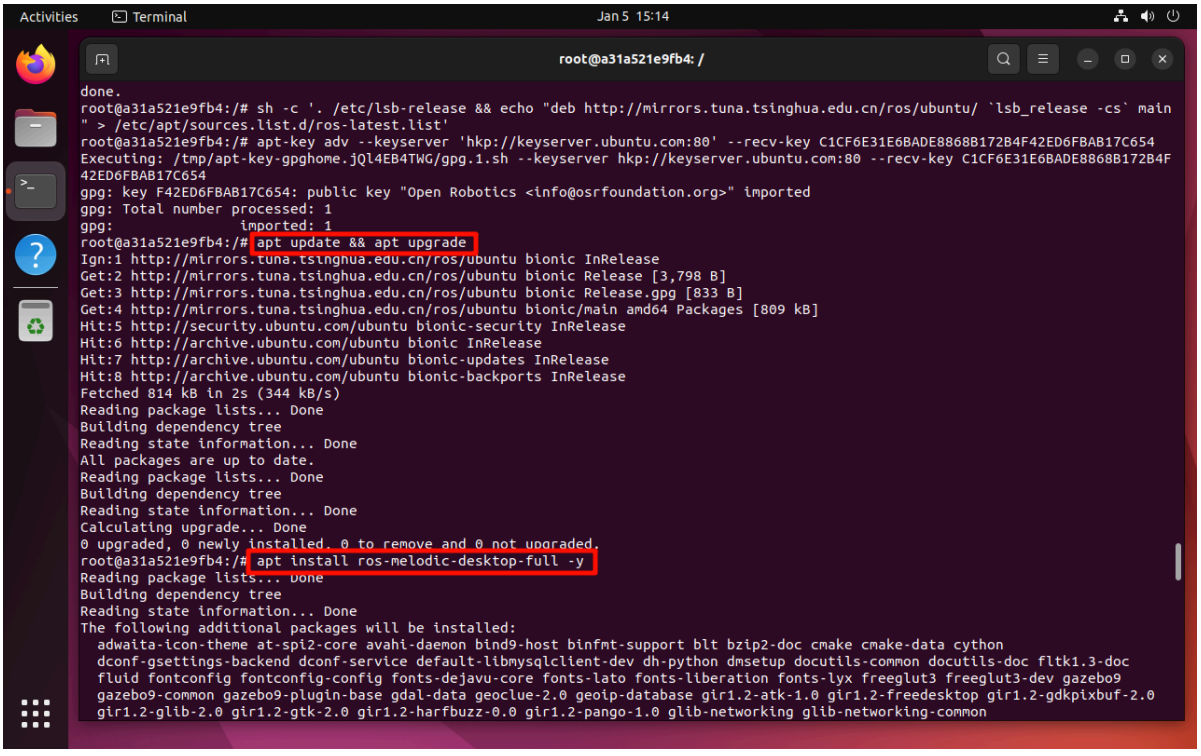


2.5. Install the ROS 1 Desktop Version

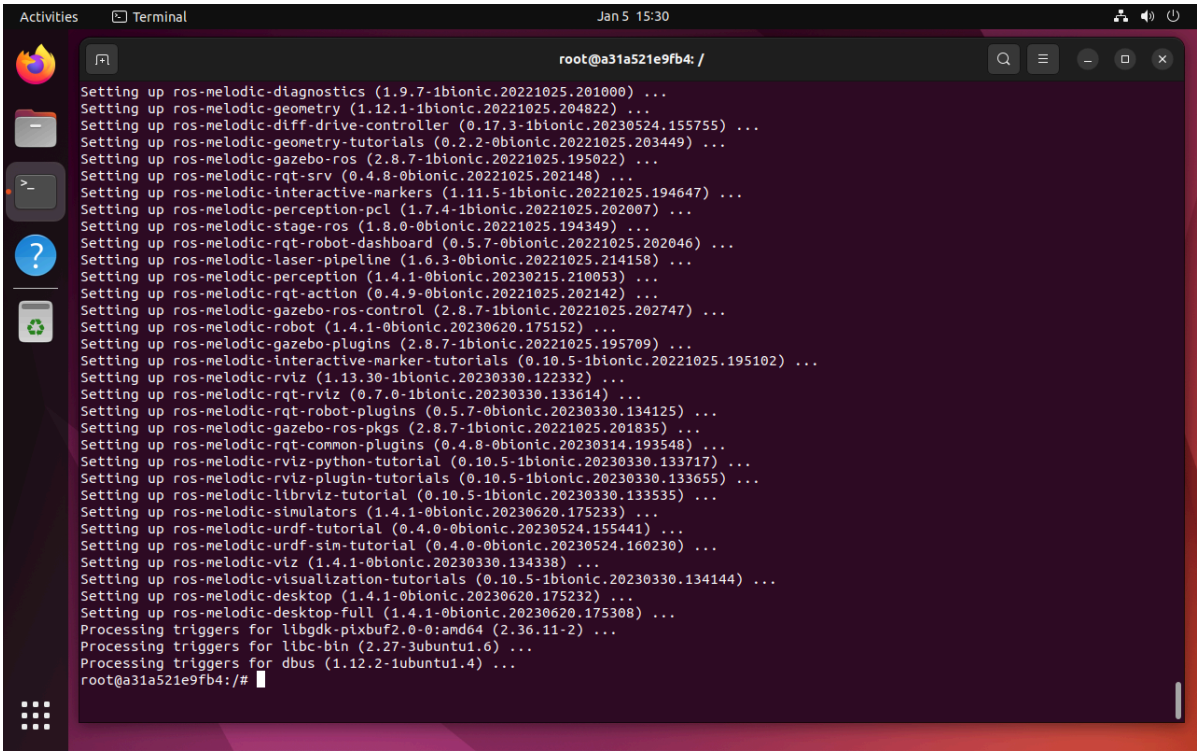
The ROS desktop version includes ROS 1 core functions, common tools, and libraries:

```
apt update && apt upgrade
```

```
apt install ros-melodic-desktop-full -y
```

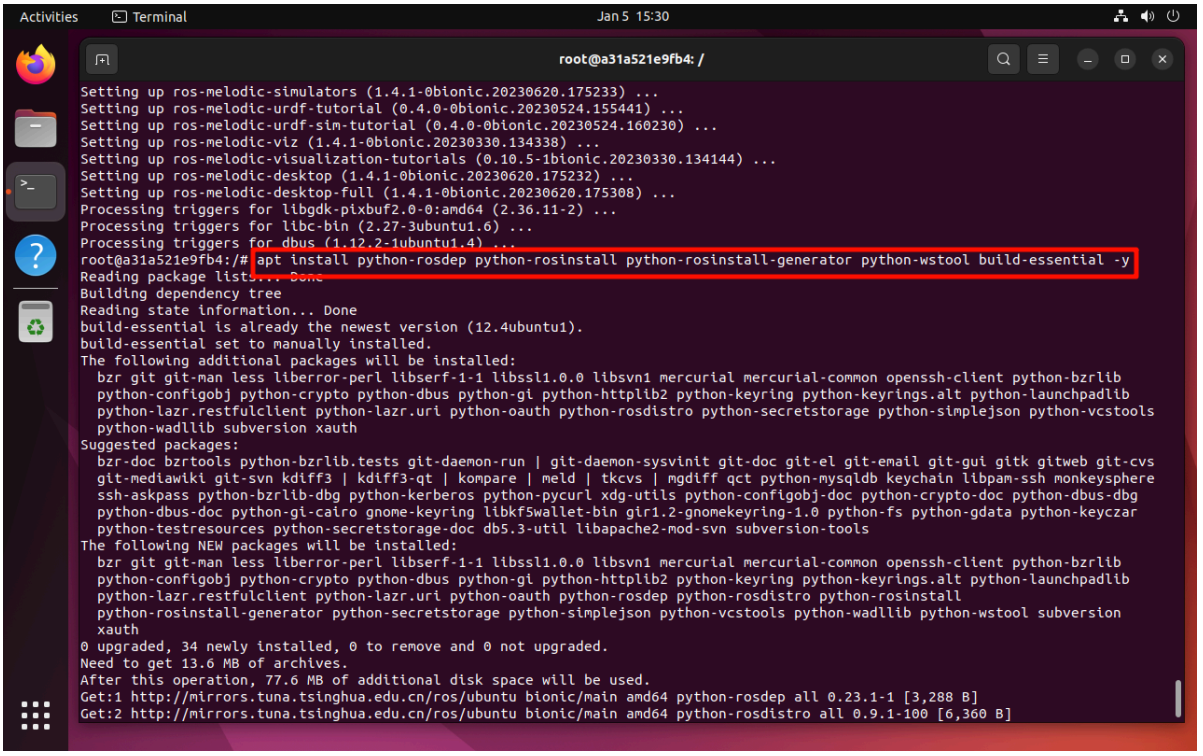


In the middle, you need to select the region and city by yourself: not demonstrated here



2.6. Install ROS 1 Dependencies

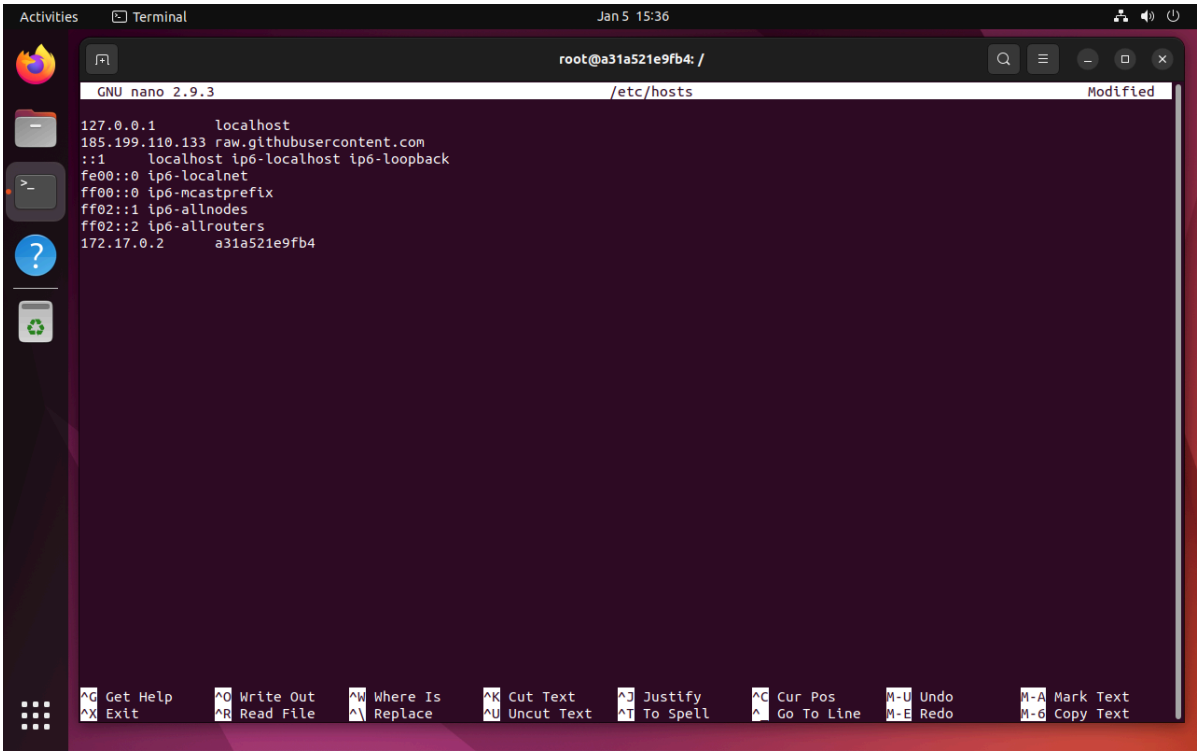
```
apt install python-rosdep python-rosinstall python-rosinstall-generator python-wstool build-essential -y
```



2.7. rosdep Initialization

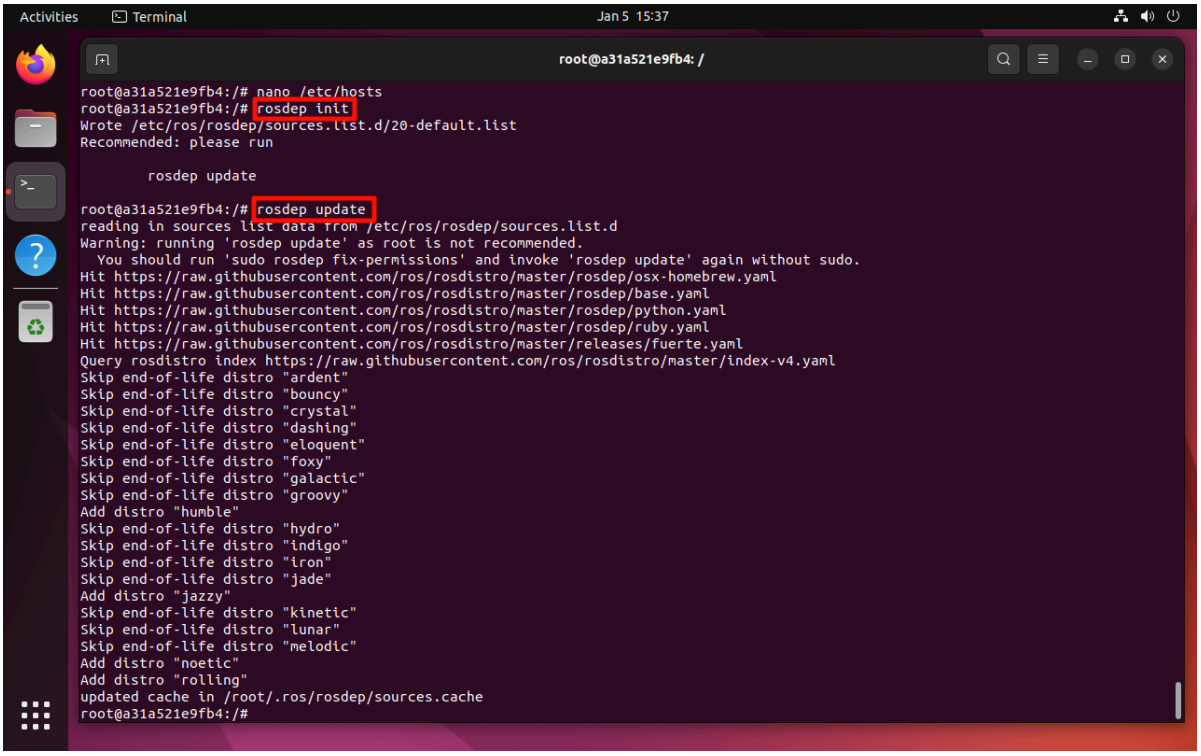
```
rosdep init
rosdep update
```

If `rosdep init` reports a download error, first check the DNS, network proxy, and HTTPS connectivity to `raw.githubusercontent.com` of the container. Do not write a fixed IP obtained from a third party into `/etc/hosts`; GitHub addresses may change, and a fixed resolution causes subsequent connection errors.



Run again:

```
rosdep init
rosdep update
```



2.8. Set Environment Variables

2.8.1. Temporary Setting

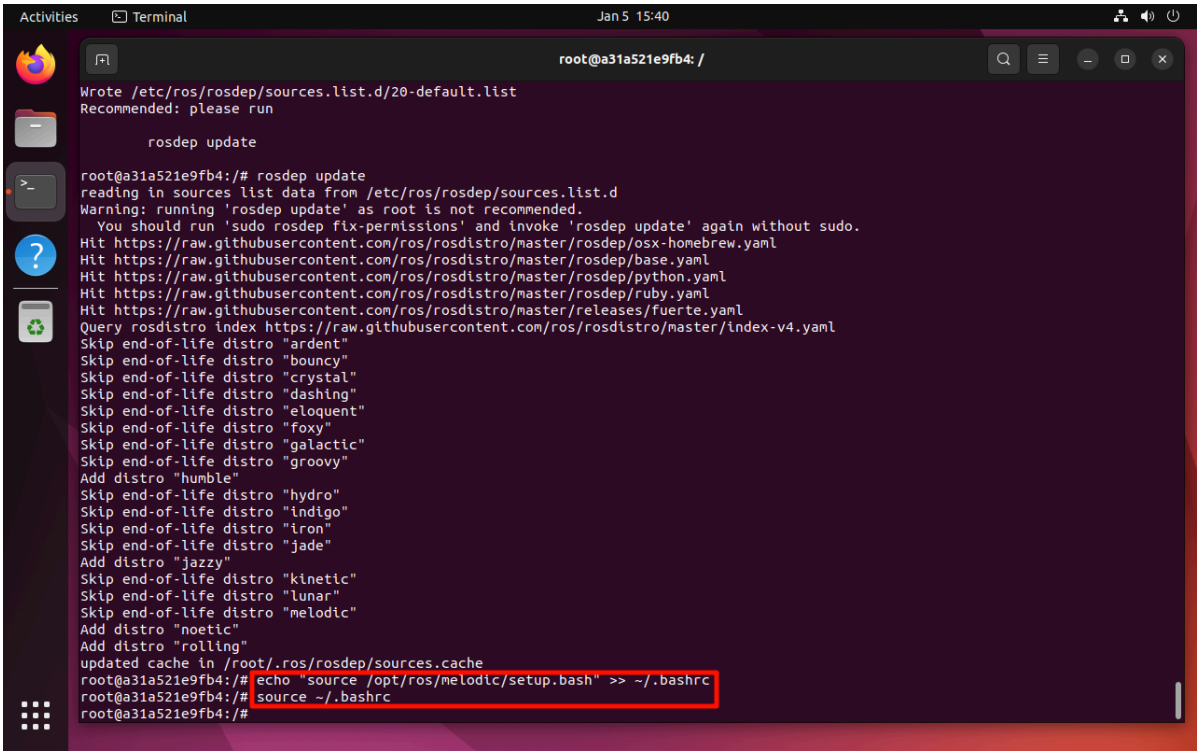
Refresh the environment variables of the current terminal:

```
source /opt/ros/melodic/setup.bash
```

2.8.2. Automatic Setting (Recommended)

Refresh the environment variables automatically on terminal startup (by sourcing .bashrc).

```
echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

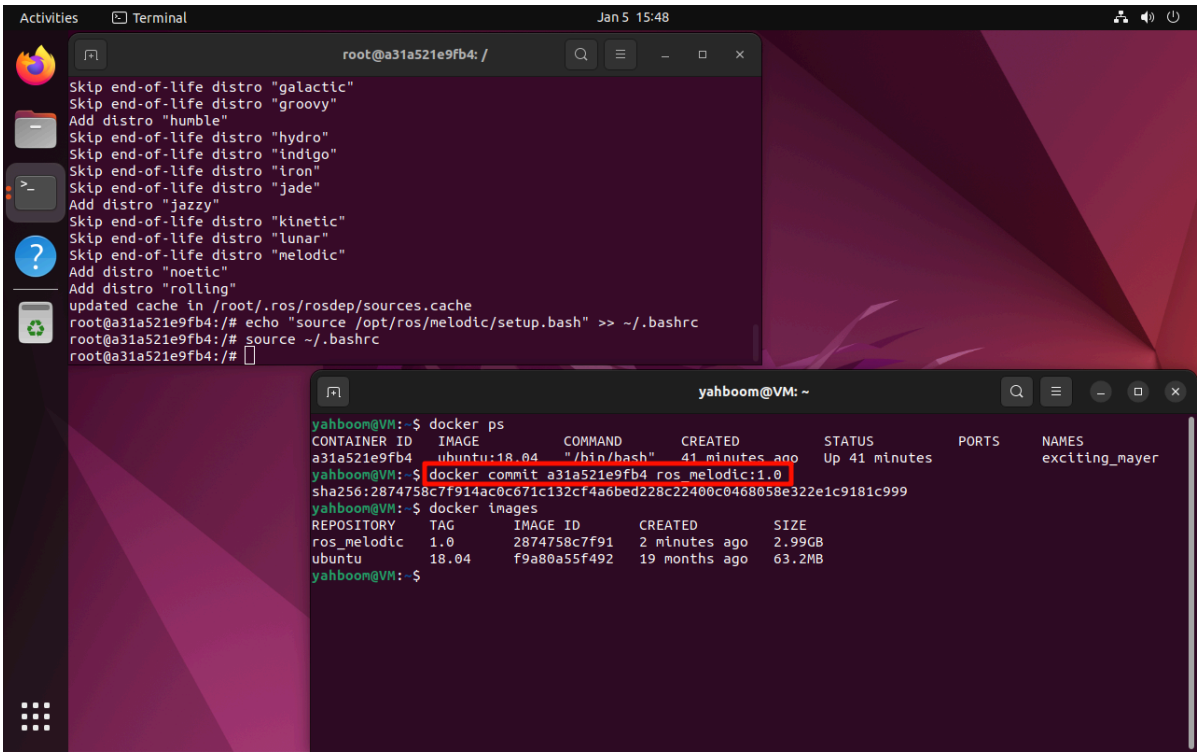


3. Commit the Image

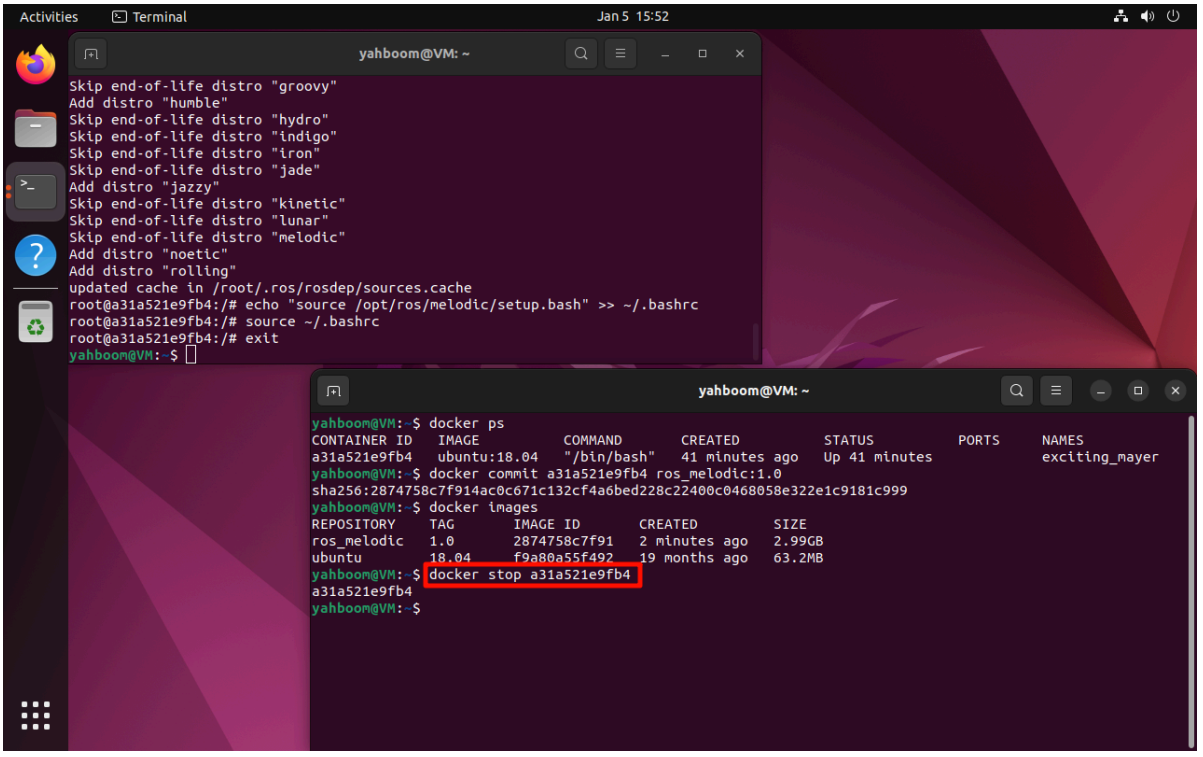
Only by committing a new image locally can the previously built ROS environment be saved; otherwise, the next time the docker image is started, there is still no ROS environment in the container.

Note: Modify according to the actual `CONTAINER ID`

```
docker commit a31a521e9fb4 ros_melodic:1.0
```



After saving, you can close the `ubuntu:18.04` container:



4. Verify the Image/ROS

4.1. Start the Image

Start the `ros_melodic:1.0` image in interactive mode: since the turtle case needs to be displayed in the graphical interface, Docker needs to be allowed to access the X Display

```
xhost +si:localuser:root
```

Revoke the authorization after the container exits:

```
xhost -si:localuser:root
```

```
docker run -it -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix
ros_melodic:1.0 /bin/bash
```

4.2. Verify the ROS Environment

Use the turtle case to test the ROS environment: open three terminals to enter the same container, and then run the following commands respectively

```
roscore
```

```
rosrun turtlesim turtlesim_node
```

```
rosrun turtlesim turtle_teleop_key
```

Note: The mouse must stay in the third command window to control the turtle

